

Jamii Savings Integration Platform — Solution Design Document

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1. Document control and design summary

Field	Value
Version	1.2 (Final draft)
Author	Emmanuel
Implements	BRD v1.2, Jamii Savings Integration Platform
Status	Final draft

Three MI integrations sit behind one APIM gateway, and one Jenkins pipeline deploys all of them to a docker-compose stack. Every API returns the same error contract and masks sensitive data in its logs.

Key decisions

Area	Decision
Balance backend	H2 2.x in server (TCP) mode in its own container, exposed through an MI Data Service with parameterised queries
Customer backend	Beeceptor mock endpoint /customers/{id} with bank-shaped records (national ID, +254 phone, KYC status) and rule-driven 404, 5xx and delay; WireMock mirror used in CI
Loan backend	DNE Online Calculator SOAP service (Divide computes debt-to-income %), with a WireMock stand-in for demos
Security	OAuth2 for Balance and Loan; API Key for Customer
Throttling	Balance and Customer on Gold; Loan on a custom LoanCheck-10PerMin tier
Gateway policy	Custom APIM operation policy on the Balance API: correlation ID and channel header injection
Packaging	Every component ships as a versioned Docker image (MI with CARs, APIM, H2, WireMock, ELK, Jenkins agent), pushed to a local registry
Release safety	Validate everything first, deploy, smoke test, roll back on any failure
Edge	nginx as the single public entry, TLS with a wildcard certificate, one subdomain per service under jamii.daqiqtech.co.ke (section 8.3); stretch goal

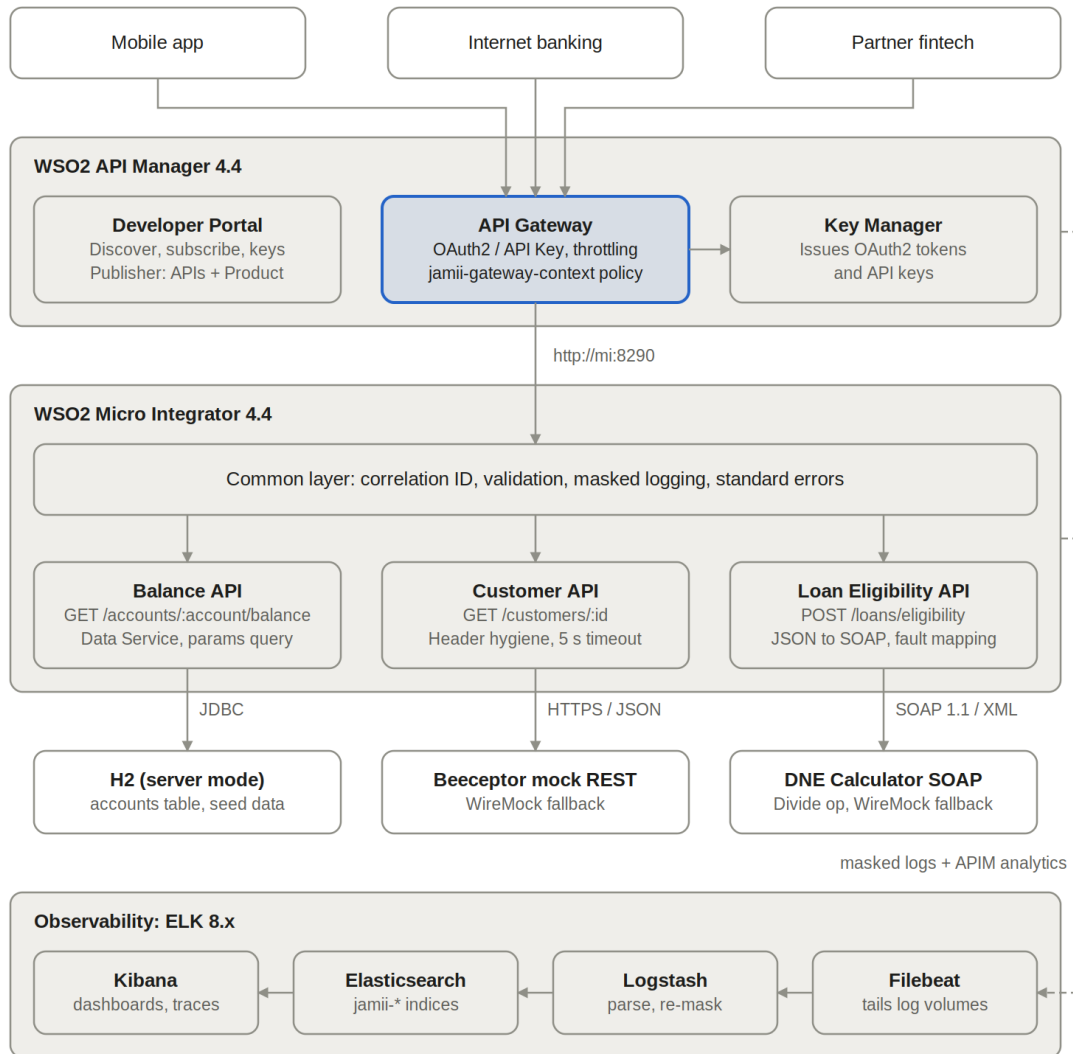
Observability: MI and APIM logs flow through Filebeat and Logstash into Elasticsearch. Kibana provides dashboards and end-to-end tracing by correlation ID (section 8.2). ELK and the nginx edge are stretch goals, started only after the core pipeline is green, to hold the 12 Oct submission.

Deployment model: Option C, self-managed. APIM and MI both run as self-managed containers, not on WSO2's managed cloud. Their images are built with docker compose build from product distributions downloaded from wso2.com, using our own Dockerfiles. The team therefore owns hosting, patching, keystores, credentials and backups (section 8.1).

2. Architecture overview

Consumers only ever talk to the APIM gateway, through nginx at api.jamii.daqiqtech.co.ke. MI is an internal layer that normalises three very different backends behind one contract.

Every consumer call passes nginx and the APIM gateway before it reaches MI



Logical architecture: consumers, APIM, MI, backends and ELK

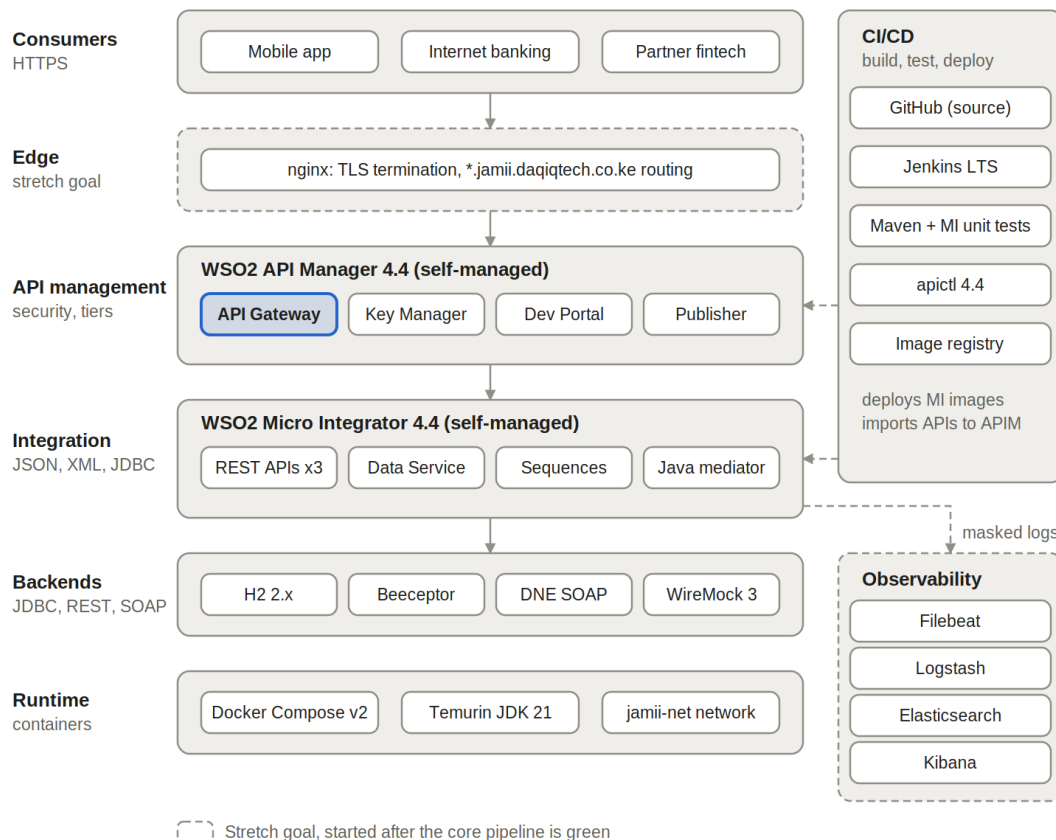
The gateway handles identity, quotas and edge policy. MI handles validation, transformation, masking and error mapping. The backends never see consumer credentials. APIM and MI both ship masked logs to the ELK stack (dashed line).

2.1 High-level design and technologies

The HLD shows the solution as layers, each with the technology that implements it. Requests flow down through the layers. CI/CD deploys into the two WSO2 layers from the side, and observability collects masked logs from the side.

One gateway, one integration layer, one pipeline

Jamii Savings high-level design, with the technology used at each layer



High-level design: layers and technologies

3. Technology stack

These are the target versions. On Day 1, confirm the latest matching MI/APIM/apictl release and pin it in .env.

Layer	Tool	Version (pin)	Purpose
Integration	WSO2 Micro Integrator	4.4.0	Mediation, data services
Tooling	WSO2 MI VS Code extension + Maven 3.9	latest / 3.9.x	Project, CAR build, unit tests
API management	WSO2 API Manager (all-in-one)	4.4.0	Publisher, Dev Portal, Gateway, Key Manager
CLI	apictl	4.4.x	Export/import APIs and products per environment
Database	H2	2.x, server (TCP) mode	Accounts data
Mock	WireMock	3.x	SOAP/REST fallback and fault injection
Log shipping	Filebeat	8.x	Tails MI and APIM log volumes
Log processing	Logstash	8.x	Parse, enrich, re-mask, route to indices
Search and storage	Elasticsearch	8.x, single node	Log and APIM analytics indices
Dashboards	Kibana	8.x	Operations dashboard, trace by correlation ID
Containers	Docker, Compose v2, registry:2	latest	Build, tag, store and run every image
CI/CD	Jenkins LTS (custom image: JDK 21, Maven, apictl, curl,	LTS	Declarative pipeline

Layer	Tool	Version (pin)	Purpose
	jq, Docker CLI)		
Tests	MI unit testing framework, gateway-check.sh (curl)	latest	Functional, contract, schema tests
Edge	nginx	stable (pinned)	TLS termination, host-based routing, IP allowlists

All four Elastic components are pinned to the same minor version.

4. Component design per API

Each API is an MI REST API artifact that calls shared sequences for validation, logging and errors. Only the backend call and the transformation differ.

4.1 Balance API (/accounts/{accountNumber}/balance)

Artifact	Name	Responsibility
Datasource	JamiiAccountsDS	JDBC pool to H2 at jdbc:h2:tcp://h2:9092/jamii (H2 driver jar baked into the MI image); credentials from env vars
Data Service	AccountsDataService	Query getAccountByNumber: SELECT account_number, status, balance, currency FROM accounts WHERE account_number = ?
REST API	BalanceAPI	Validates path, calls data service, builds envelope
Fault sequence	Common_FaultSeq	Maps SQL/connection errors to 503 DB_UNAVAILABLE

Flow: validate `^[0-9]{10,16}$` (else 400), call the data service. Zero rows returns 404 ACCOUNT_NOT_FOUND. One row is transformed with a PayloadFactory into the BRD envelope, with requestId = correlation ID and timestamp = UTC now. A JDBC or connection error triggers the fault sequence and returns 503.

4.2 Customer API (/customers/{customerId})

Artifact	Name	Responsibility
HTTP endpoint	CustomerBackendEP	https://jamii-customers.free.beeceptor.com/customers/{uri.var.customerId} (endpoint name confirmed at setup); timeout 5000 ms, action fault
REST API	CustomerAPI	Header manipulation, pass-through body, 404 mapping
Sequence	Common_HeaderHygiene	Add/propagate X-Correlation-ID; strip X-Internal-*, Server, X-Powered-By, Set-Cookie

Flow: validate `^[0-9]{1,10}$` (else 400), apply header hygiene and call the endpoint. A 200 body is passed through and the X-Correlation-ID response header is set. A backend 404 returns 404 CUSTOMER_NOT_FOUND. A timeout returns 504 BACKEND_TIMEOUT. Connection refused or DNS failure triggers the fault sequence and returns 503 BACKEND_UNAVAILABLE. The backend URL is a config property, so it can be switched to WireMock without a code change.

Beeceptor rules. The mock returns records shaped like a Kenyan bank's customer file, so masking is exercised on national ID and phone, not just names.

Request	Beeceptor response	Purpose
GET /customers/1001 to 1005	200 with customerId, fullName, nationalId, phone (+254), email, kycStatus, branch	Happy path and PII masking
GET /customers/9999	404	CUSTOMER_NOT_FOUND mapping

Request	Beeceptor response	Purpose
GET /customers/5000	500	502 BAD_GATEWAY mapping
GET /customers/5004	200 after a 7 s delay	Exceeds the 5 s timeout, so 504 BACKEND_TIMEOUT

The rules are exported to `mocks/beeceptor-rules.json` and mirrored one-for-one in the WireMock stubs. The pipeline and gateway checks always use WireMock, so they never consume the Beeceptor free-tier request quota; Beeceptor is used for the live demo and recording.

4.3 Loan Eligibility API (POST /loans/eligibility)

Request:

```
{ "customerId": "7", "monthlyIncome": 120000, "existingMonthlyDebt": 15000,
  "requestedMonthlyInstalment": 25000 }
```

Mapping to SOAP Divide(intA, intB):

- `intA` = (existingMonthlyDebt + requestedMonthlyInstalment) × 100, computed in MI
- `intB` = monthlyIncome
- The result is the debt-to-income ratio as a whole percent; eligible when it is 40 or less.

Response:

```
{ "customerId": "7", "decision": "ELIGIBLE", "eligible": true, "debtToIncomePercent": 33,
  "maxAllowedPercent": 40, "requestId": "...", "timestamp": "..." }
```

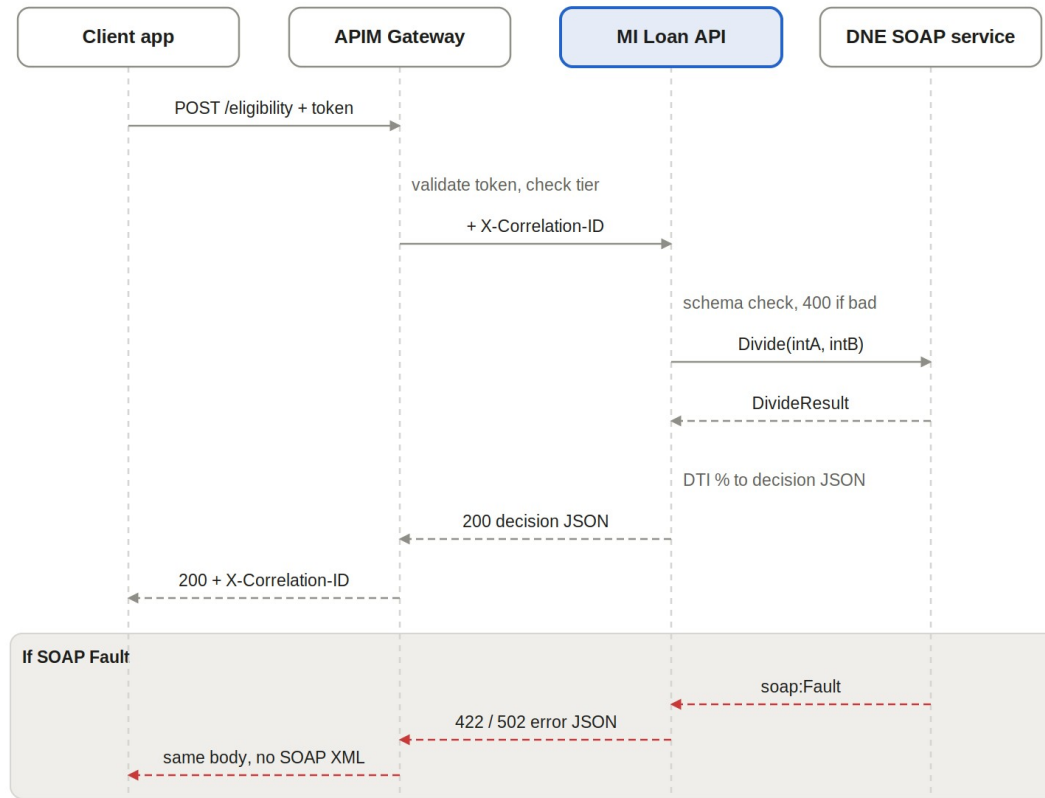
Why this mapping: a debt-to-income ratio is a real lending affordability test. The calculator service performs the one arithmetic step, which keeps the SOAP call meaningful rather than decorative. Income of 0 passes JSON validation but makes the SOAP service throw a divide-by-zero fault. That gives a genuine SOAP Fault to demonstrate.

SOAP outcome	REST response
DivideResult returned	200 with decision
Fault: divide by zero	422 ELIGIBILITY_REJECTED, "Monthly income must be greater than zero"
Any other fault	502 BAD_GATEWAY (fault string logged, never returned)
Timeout over 10 s	504 BACKEND_TIMEOUT
Unreachable	503 BACKEND_UNAVAILABLE

5. Loan Eligibility request flow

This is the most complex flow, because it crosses protocols twice: JSON to SOAP, then SOAP back to JSON.

A SOAP fault never reaches the client as XML



Loan Eligibility sequence: happy path and SOAP fault path

Solid arrows are requests and dashed arrows are responses. Red marks the fault path, where MI's fault sequence turns the SOAP fault into the standard error JSON before anything is returned.

6. Cross-cutting design

Shared logic lives in four reusable sequences and one template, so each API stays thin and every API behaves the same way.

Artifact	Type	What it does
Common_InSeq	Sequence	Reads or generates X-Correlation-ID (UUID v4), stores it as property correlationId, records start time
Common_MaskedLog	Sequence template	Logs a stage (REQUEST_IN, BACKEND_CALL, RESPONSE_OUT) with masked fields; params: stage, api, extra
Common_ErrorResponse	Sequence template	Builds the standard error JSON from params (httpStatus, code, message, details), sets status and correlation header, responds
Common_FaultSeq	Fault sequence	Catches unhandled errors, classifies them by ERROR_CODE (timeout 101504, connection 101503, SQL) and calls Common_ErrorResponse
Common_HeaderHygiene	Sequence	Strips internal and server-identifying headers inbound and outbound

Validation: each API validates input before any backend call. Path parameters use regex. The loan body uses a JSON Schema (loan-request.schema.json, stored as a registry resource) via the Validate mediator. A failure returns 400 VALIDATION_ERROR with field-level details. Malformed JSON is caught in the fault sequence and also returns 400, never 500.

Masking rules: masking happens in a small Class Mediator (MaskingMediator), so the logic is unit-testable Java rather than repeated expressions.

Field	Rule	Example
accountNumber	keep last 4	*****7890
name, username, email, phone, address	full mask	***
monthlyIncome, debts, instalment	omitted from logs	n/a

Logging format: one line per stage, key=value, for example:

```
stage=REQUEST_IN api=BalanceAPI correlationId=5b1f... account=*****7890 method=GET
stage=BACKEND_CALL api=BalanceAPI correlationId=5b1f... backend=AccountsDataService
stage=RESPONSE_OUT api=BalanceAPI correlationId=5b1f... httpStatus=200 latencyMs=42
```

7. API Manager design

All three APIs are published from apictl projects in Git, never hand-edited in the Publisher. Git is the source of truth.

API	Context / version	Security	Subscription tier	Backend (MI)
Jamii Account Balance	/jamii/accounts/v1	OAuth2	Gold	http://mi:8290/accounts
Jamii Customer Profile	/jamii/customers/v1	API Key	Gold	http://mi:8290/customers
Jamii Loan Eligibility	/jamii/loans/v1	OAuth2	LoanCheck-10PerMin (custom)	http://mi:8290/loans

Tier reason (for README): Loan eligibility gets a tighter custom tier because each call fans out to a slow, external SOAP dependency with no SLA, and the tier protects it from overload.

Security reason: OAuth2 suits user-context and financial data (Balance, Loan). API Key shows the simpler server-to-server option for partner lookups (Customer). Using both shows the trade-off.

Custom mediation policy: jamii-gateway-context is an APIM 4.x operation policy (.j2 + .yaml) applied on the request flow of the Balance API. It injects X-Correlation-ID if absent, so tracing starts at the edge. It adds X-Channel from the calling application's name and removes the Authorization header before forwarding, so tokens never reach MI.

OpenAPI: hand-written OAS 3.0 per API, including path parameter patterns, request schema, the success schema, and every error response (400, 401, 404, 422, 429, 502, 503, 504) referencing one shared ErrorResponse component.

API Product: Jamii Retail Banking bundles all three, with OAuth2 and the Gold tier. One subscription gives a channel app (for example the mobile app) access to all three. Partners can still subscribe to individual APIs.

Developer Portal onboarding (consumer steps)

1. Sign in at <https://portal.dev.jamii.daqiqtech.co.ke/devportal> and browse or search the jamii tag.
2. Read the API docs and try the OpenAPI console.
3. Create an application (for example jamii-mobile) and choose a business plan.
4. Subscribe the application to the Jamii Retail Banking product, or to individual APIs.
5. Under Production Keys, generate client credentials (OAuth2) or an API key.
6. Call the gateway at <https://api.dev.jamii.daqiqtech.co.ke/jamii/...> with Authorization: Bearer <token> or the apikey header.

Until the nginx stretch is delivered, the same steps run locally against <https://localhost:9443/devportal> and the gateway at <https://localhost:8243>.

8. Deployment architecture

One docker compose up -d starts the whole dev environment on a single bridge network, jamii-net. Only APIM's gateway and portal ports are meant for consumers; MI is reachable directly only for debugging.

Service	Image	Built from	Baked into the image	Host ports	Compose profile
h2	jamii/h2:\${TAG}	eclipse-temurin:21-jre + H2 jar	schema.sql + seed.sql, run by RunScript before the TCP server starts	9092, 8082 (console, dev only)	core
mi	jamii/mi:\${TAG}	eclipse-temurin:21-jre + wso2mi-4.4.0.zip	3 CARs, H2 driver, MaskingMediator jar, log4j2 config	8290, 8253, 9164	core
apim	jamii/apim:\${TAG}	eclipse-temurin:21-jre + wso2am-4.4.0.zip	deployment.toml (ELK analytics on), custom policy, keystores	9443, 8243, 8280	core
wiremock	jamii/wiremock:\${TAG}	wiremock/wiremock:3.x	SOAP and REST stubs, fault scenarios	8089	core
filebeat	jamii/filebeat:\${TAG}	elastic/filebeat:8.x	filebeat.yml (MI and APIM inputs)	none	observability
logstash	jamii/logstash:\${TAG}	logstash:8.x	Pipeline: kv/json parse, re-mask, index routing	5044	observability
elasticsearch	stock elasticsearch:8.x	n/a	Config via env, 1 GB heap	9200	observability
kibana + kibana-setup	stock kibana:8.x + jamii/kibana-setup:\${TAG}	curl image	dashboards.ndjson imported once at start	5601	observability
jenkins	jamii/jenkins:\${TAG}	jenkins/jenkins:lts-jdk21	Maven, apictl, curl, jq, Docker CLI, plugins list	8080	ci
registry	stock registry:2	n/a	n/a	5000	ci
nginx	jamii/nginx:\${TAG}	nginx:stable-alpine	One server block per subdomain, TLS settings, IP allowlists (certificates mounted, not baked)	80, 443	core

Images carry platform configuration; API definitions do not. The three APIs and the product are imported into APIM by apictl at deploy time. That way the same jamii/apim image is reused in every environment, and only api_params.yaml changes.

The APIM and MI Dockerfiles unpack the downloaded zips from dist/, which is gitignored because the files are large. Each zip's SHA-256 is checked against dist/checksums.txt before the build. The containers run as a non-root wso2 user. Jenkins reads the same zips from a mounted dist volume, so CI builds exactly what runs locally.

docker compose --profile core up -d is enough for the API demo. Add --profile observability for ELK, and --profile ci for Jenkins and the registry.

8.1 Self-managed responsibilities (Option C)

Because both APIM and MI are self-managed, every operational concern below is ours. The table shows how this project handles each one, and what would change for production.

Responsibility	This project (dev)	Production
Hosting and scaling	docker-compose, one node per	Kubernetes with WSO2 Helm charts;

Responsibility	This project (dev)	Production
	component	2+ gateway and MI replicas
Product updates	Downloaded GA distributions, version pinned, checksum verified	WSO2 Updates (subscription) applied by rebuilding images
APIM internal databases	Built-in H2 (APIM default)	External RDBMS for apim_db and shared_db
Keystores and TLS	Default keystores replaced with project self-signed certs	CA-signed certificates, secure vault for secrets
Admin credentials	admin/admin changed via env vars and Jenkins credentials	IdP-backed users, least-privilege roles
Backups	Not needed (disposable stack)	Scheduled DB and config backups, tested restore
Monitoring	ELK dashboards	ELK plus alerting and on-call

Environments: each environment is a set of config values, not separate code.

Setting	dev	prod (conceptual)
apictl env name	dev	prod
APIM endpoints	apim.dev.jamii.daqiqtech.co.ke	apim.jamii.daqiqtech.co.ke
MI backend URL in API params	http://mi:8290	prod MI load balancer
Backends	public services or WireMock	real core/credit systems
Deployment trigger	automatic on main	manual approval

API endpoints differ per environment through apictl api_params.yaml, so the same API archive is promoted unchanged.

8.2 Observability (ELK), stretch

Logs are masked at the source in MI, shipped by Filebeat, parsed and re-checked by Logstash, stored in Elasticsearch, and explored in Kibana. ELK never receives an unmasked account number.

1. MI writes its masked key=value mediation logs and access logs to a shared volume. APIM, with analytics type elk in deployment.toml, writes gateway events to apim_metrics.log.
2. Filebeat tails both volumes and tags each event with its source (mi or apim).
3. Logstash parses MI lines with the kv filter and APIM events as JSON. It sets @timestamp, re-masks anything matching \b\d{10,16}\b to its last 4 digits, and routes events to jamii-mi-* or jamii-apim-analytics-.*
4. Elasticsearch runs as a single node with a 1 GB heap; an ILM policy deletes indices after 7 days.
5. kibana-setup imports data views and the dashboard on first start.

Dashboard panel	Source	Question it answers
Requests per API per minute	APIM analytics	Who is calling what, and how much
Error rate by error code	MI logs	What is failing, and why
p95 latency per API and backend	MI latencyMs	Which backend is slow
Throttled (429) calls by application	APIM analytics	Is a tier too tight
Trace by correlation ID (saved search)	MI + APIM	What happened to this one request

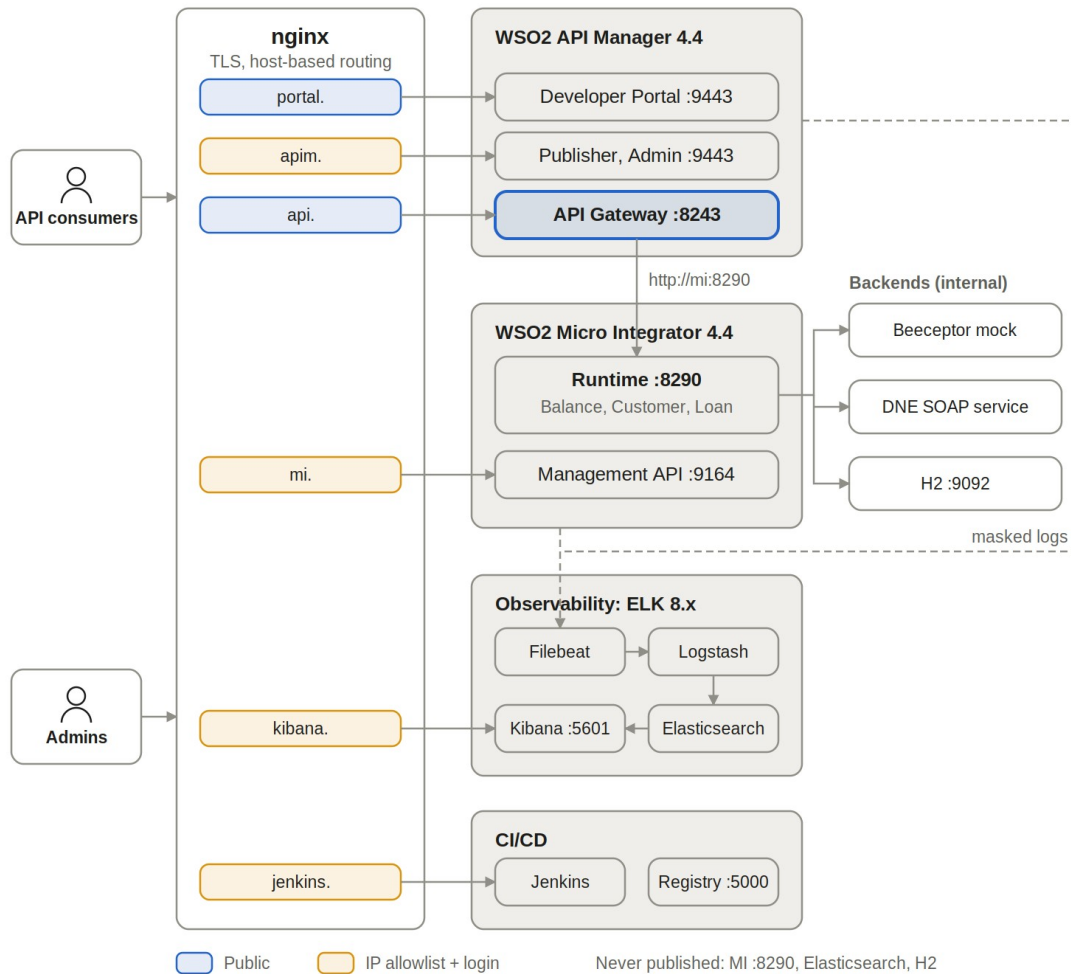
Resource budget: APIM 2 GB, Elasticsearch 1.5 GB, Logstash 1 GB (512 MB heap), MI 1 GB, Kibana 0.7 GB, Jenkins 1 GB, others under 0.5 GB. That is about 8 GB, so a 16 GB machine is the minimum. If memory is tight, Logstash can be dropped and Filebeat can write to an Elasticsearch ingest pipeline instead.

8.3 Edge topology: nginx and subdomains, stretch

nginx is the single public entry point. It terminates TLS with one wildcard certificate and routes by hostname, so no WSO2, Elastic or Jenkins port is ever exposed directly.

nginx is the only public entry, and each service gets its own subdomain

One wildcard TLS certificate: *.jamii.daqiqtech.co.ke (dev: *.dev.jamii.daqiqtech.co.ke)



Edge topology: nginx, subdomains and internal services

Subdomain (prod)	Dev equivalent	Routes to	Exposure
api.jamii.daqiqtech.co.ke	api.dev.jamii.daqiqtech.co.ke	apim:8243 (gateway, incl. /oauth2/token)	Public
portal.jamii.daqiqtech.co.ke	portal.dev.jamii.daqiqtech.co.ke	apim:9443 /devportal	Public
apim.jamii.daqiqtech.co.ke	apim.dev.jamii.daqiqtech.co.ke	apim:9443 /publisher, /admin, /carbon	IP allowlist + login (Jenkins IP allowed for apictl)
mi.jamii.daqiqtech.co.ke	mi.dev.jamii.daqiqtech.co.ke	mi:9164 management API only	IP allowlist + login
kibana.jamii.daqiqtech.co.ke	kibana.dev.jamii.daqiqtech.co.ke	kibana:5601	IP allowlist + Elastic login
jenkins.jamii.daqiqtech.co.ke	jenkins.dev.jamii.daqiqtech.co.ke	jenkins:8080	IP allowlist + Jenkins login; webhook path open

MI's runtime (8290), Elasticsearch, H2 and the registry stay on the internal Docker network. Publishing MI's runtime would let callers bypass the gateway's security and throttling.

nginx and WSO2 configuration notes

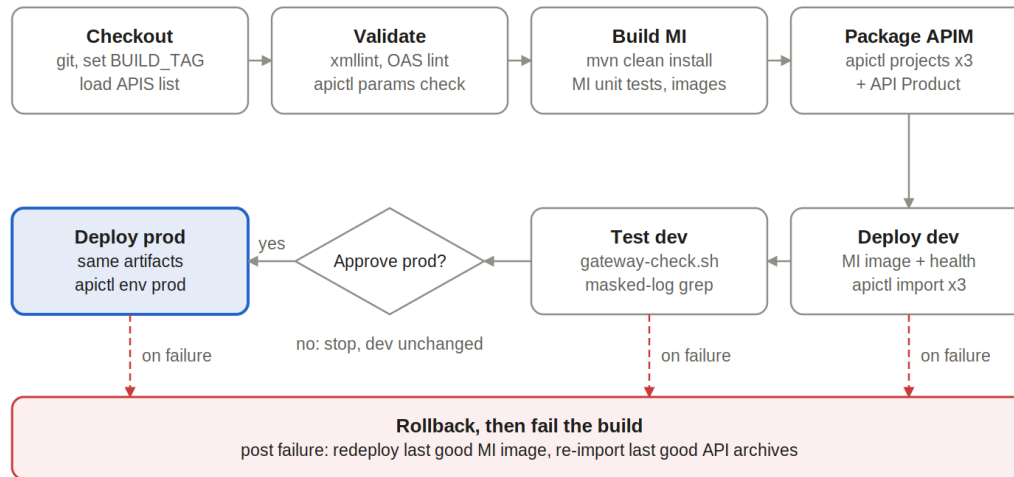
- APIM must know its public hostnames. Set the server hostname, proxy port 443, and the gateway and Developer Portal URLs in deployment.toml; otherwise logins and redirects point at internal ports. Verify the exact keys against the 4.4 docs.
- WSO2 listens on HTTPS internally with self-signed certificates, so nginx uses proxy_pass https://apim:9443 and proxy_ssl_verify off. This is acceptable on the internal network for dev; prod uses trusted certificates.

- Raise `client_max_body_size` (for example to 50m) on apim. so apictl API imports are not rejected.
- Set Kibana `server.publicBaseUrl` and the Jenkins URL to their public hostnames.
- Locally, there is no real DNS: use mkcert for the wildcard certificate and /etc/hosts entries pointing the six hostnames to 127.0.0.1.

9. CI/CD pipeline design

One declarative Jenkinsfile builds and deploys all three APIs together. They are released as one unit or not at all. Every image is tagged with the build number and pushed to the local registry, so dev and prod run identical images.

Any failed stage rolls back, and prod deploys only after approval



Jenkins pipeline: stages, approval gate and rollback path

How partial deploys are prevented

1. Everything is validated and packaged before anything is deployed, so a broken spec fails the run while all environments are untouched.
2. Before deploying, the pipeline records the current MI image tag and exports the current API archives (apictl export api) as rollback artifacts.
3. Deploy dev runs the MI image swap, waits for the MI health check, then imports the three APIs in a loop with set -e. The first failure stops the loop.
4. `post { failure { sh 'scripts/rollback.sh' } }` restores the previous image and re-imports the saved archives, then the build is marked FAILED.
5. Prod uses the same artifacts (no rebuild), with apictl env prod and an input approval step. In this assignment it is wired as a dry run.

Credentials (APIM admin, DB password) come from the Jenkins credentials store via withCredentials and are never echoed.

10. Repository structure

```

jamii-savings-integration/
├── README.md
├── Jenkinsfile
├── docker-compose.yml      # profiles: core, observability, ci
├── .env.example           # TAG, registry, passwords (no real secrets)
├── dist/                  # downloaded WSO2 zips + checksums.txt (zips gitignored)
├── mocks/                 # beeceptor-rules.json (exported Beeceptor rules)
├── db/
│   ├── schema.sql
│   └── seed.sql

```

```

mi/                                # MI multi-module Maven project
├── pom.xml
├── common/                        # shared sequences, templates, MaskingMediator
├── balance-api/                  # BalanceAPI, AccountsDataService, datasource
├── customer-api/                 # CustomerAPI, CustomerBackendEP
├── loan-api/                     # LoanEligibilityAPI, SOAP endpoint, schema
├── composite/                   # CAR packaging
├── apim/
│   ├── apis/
│   │   ├── JamiiAccountBalance-v1/ # apictl project (api.yaml, OAS, policies)
│   │   ├── JamiiCustomerProfile-v1/
│   │   └── JamiiLoanEligibility-v1/
│   ├── products/JamiiRetailBanking/
│   ├── policies/jamii-gateway-context/
│   ├── throttling/LoanCheck-10PerMin.json
│   └── params/                  # api_params.yaml per environment
├── docker/                      # one Dockerfile + config per image
├── h2/
├── mi/
├── apim/                        # deployment.toml (ELK analytics)
├── wiremock/                   # mappings + __files
├── filebeat/
├── logstash/                   # pipeline/jamii.conf
├── nginx/                     # conf.d per subdomain; certs gitignored
├── kibana-setup/               # dashboards.ndjson
├── jenkins/                   # plugins.txt, tools
├── tests/
│   ├── gateway/               # gateway-check.sh settings per environment
│   └── schemas/error-response.schema.json
├── scripts/                    # build-images.sh, deploy-mi.sh, deploy-apim.sh, gateway-check.sh,
rollback.sh
└── docs/                      # BRD, design doc exports, diagrams, deck

```

The Jenkinsfile loops over a single list, APIS = ['balance','customer','loan']. Adding a fourth API means adding a folder and one list entry, with no new pipeline code.

11. Testing strategy

Tests run in the pipeline in order of cost: fast checks first, full gateway tests last.

Level	Tool	What it proves	Pipeline stage
Static	xmllint, OAS lint (spectral), apictl validation	Artifacts and specs are well formed	Validate
Unit	JUnit for MaskingMediator	Masking rules work in isolation	Build
API behaviour	MI unit testing framework (test suites in each MI module, mock services for the REST and SOAP endpoints, H2 container for Balance)	Each API: happy path, 400, 404, 422/502 SOAP fault mapping, backend down	Build (mvn clean install)
Contract	JSON-path assertions in every MI error test (\$error.code, \$error.httpStatus, \$requestId, \$timestamp)	Every error body has the same shape	Build
Gateway (end to end)	scripts/gateway-check.sh (curl, exit codes) via APIM	401 without token, 200 with token, 429 over tier, product subscription works	Test (dev)
Security/logging	grep on MI logs	No full account number or	Test (dev)

Level	Tool	What it proves	Pipeline stage
		PII leaked, in log files or in Elasticsearch	

The MI unit tests need an MI runtime in CI. The Jenkins agent unpacks the MI zip from the mounted dist/ volume and points the unit test plugin at it. If the MI version cannot assert on HTTP status codes directly, tests assert on \$.error.httpStatus in the body.

Timeout scenarios, which MI mock services do not simulate well, use WireMock (fixedDelayMilliseconds over the timeout, CONNECTION_RESET_BY_PEER) and docker stop h2, so the 503 and 504 paths are tested on purpose, not by luck.

12. Design decisions, trade-offs and bonus design

ADR	Decision	Alternative rejected	Trade-off accepted
01	H2 in server mode, in its own container	H2 embedded inside MI; MySQL	Not production-grade, but light; a separate container still lets the demo stop the DB to show 503
02	Beeceptor mock endpoint for customers, mirrored in WireMock for CI	JSONPlaceholder, Reqres	Needs an account and has a free-tier request cap; chosen for bank-shaped PII and rule-driven 404, 5xx and delays
03	DNE Calculator for SOAP	CountryInfo, NumberConversion	Integer arithmetic only; the % mapping gives a meaningful affordability rule
04	WireMock fallback for both public backends	Rely on public services	More setup, but the demo and CI cannot be broken by a third party
05	Shared sequences/templates + Class Mediator for masking	Inline logic per API	One more Java module; masking stays consistent and testable
06	Correlation ID at both APIM (policy) and MI	Only one layer	Small duplication; MI stays safe when called directly
07	Every component as a versioned image in a local registry	Official WSO2 images with mounted config and hot-copied CARs	Longer builds; immutable releases, and rollback becomes "run the previous tag"
08	APIM config as apictl projects in Git, imported at deploy time	APIs baked into the APIM image	One more deploy step; one APIM image serves every environment
09	One API Product plus individual APIs	Product only	Two subscription paths to explain; fits channels and partners
10	Validate everything before deploying anything, then roll back on failure	Deploy each API as it builds	Longer pipeline; never a partial release
11	ELK with Logstash, masking at source plus re-masking in Logstash	No central logging; Filebeat straight to Elasticsearch	About 3 GB more RAM; searchable traces and a second masking safeguard
12	Elastic security on (password from .env), TLS off locally	Security disabled	Slightly more setup; mirrors how a bank would run it, while TLS is noted as a prod item
13	Self-managed WSO2 (Option C), images built from downloaded distributions	WSO2 managed cloud; official Docker images	We own patching and operations; full control of versions, config and image contents
14	nginx edge with one subdomain per service under jamii.daqiqtech.co.ke	Publish service ports directly; path-based routing on one host	One more container and certificate setup; a single TLS point, IP allowlists, no internal port exposed

Bonus A: Customer dashboard aggregation (GET /customers/{customerId}/dashboard)

- MI Scatter-Gather (or Clone + Aggregate) calls Customer and Balance in parallel, with each branch timing out at 5 s.
- Decision: **return partial data**. The response is 200 with a sections object, and a failed section is marked {"status":"UNAVAILABLE","error":{"...}}. If *both* fail, the API returns 503.
- Reason: a dashboard is read-only and still useful without one panel, so failing the whole page for one slow backend is a worse customer experience.

Bonus B: Loan application events (RabbitMQ)

- An MI RabbitMQ inbound endpoint consumes loan.applications. It validates and transforms each message, then calls /loans/eligibility and publishes the result to loan.decisions.
- Malformed or unprocessable messages are rejected without requeue and go to loan.applications.dlq through a dead-letter exchange. Transient backend errors are retried 3 times with backoff before going to the DLQ.
- How this differs from the sync APIs: there is no caller waiting, so errors cannot be returned. Instead they must be retried, parked and replayed. Idempotency (an applicationId key) also matters, because a message may be delivered more than once.